RFL™ Academy

BUILDING CHAMPIONS

SINCE 2016



visit robofunlab.com

Founder's Note

For four years, I taught robotics at a single school in Rajkot.

I trained one group of students, took them to competitions across the world, and watched what happened to them when they were given real problems and real tools. It worked. And that was the difficulty. I kept thinking about the students in the next school, and the next city, who would never get the same four years.

RFL Academy began in 2016 for that reason. One city, Rajkot, small but an engineering capital in its own right. Ten years later we work with over 50 schools across India and Saudi Arabia, and run 10 of our own robotics labs across four Indian cities.

What that has produced is not a set of certificates. It is students like Ranveer Jain at Carnegie Mellon, Kavish Contractor at WPI on a Presidential Scholarship, Ayaan Shah at UC Berkeley, and Keshav Mohta at Penn's M&T programme. It is the first Indian team to win the FIRST Impact Award, the highest honour in FRC. In 2025, FIRST named me a Woodie Flowers Finalist, an award for mentorship that only students can nominate you for. Of everything on this page, that is the one I hold closest. Along the way, Packt Publishing invited me to lead-author a book on robotics for young learners.

I will tell you what we have not finished. Every coach at RFL should be able to teach the way I teach, and several should be better than me. Building that consistently, across every branch, is the work I am most focused on and the work I am least willing to call complete.

The next ten years are simple to state. We intend to send into the world leaders, problem solvers, and logical thinkers, through our courses, our competition coaching, and whatever comes after them.

If any of this is useful to your school or your child, let us converse.



Sincerely,
Ashwin Shah
Co-founder & Director
Robofun Lab Pvt. Ltd.

Who Are We?

We started in 2016, in Rajkot, with a handful of children who stayed back after school because they wanted to build something. That is still the child we look for. Not the one who scores well, but the one who cannot leave a half finished robot alone.

Give that child a good coach and a real problem, and most of the work is done. Our job after that is mainly to keep the standard high and stay out of the way.

Ten years on, we run 10 labs across 4 cities in India, run several robotic courses for K-12 students, mentor students for STEM portfolio building and train teams for prestigious national & international robotics competitions with winning outcomes. What we learn at those competitions is what our Grade 4 children are taught the following year.

What We Do?

Robotics Labs: We operate state-of-the-art robotics labs where students engage in building and coding their own robots. Our structured curriculum is tailored for each grade, ensuring comprehensive learning and skill development.

Competition Coaching: We provide expert coaching for teams participating in prestigious national and international robotics competitions. Our guidance has consistently led students to achieve national and international championship titles.

School Collaborations: We collaborate with forward-thinking schools to deliver custom robotics and coding solutions. These partnerships enhance students' skills in these fields and prepare them for various competitive events.

Student Profile Development: We assist students in building unique STEM and Robotics portfolios by facilitating participation in recognized competitions and supporting personal projects. This helps students stand out in their academic and professional pursuits.

RFL Academy's Impactful Journey

Since 2016, RFL Academy has been a driving force in robotics education, impacting over 10000+ students through 10 labs in India and forging partnerships with more than 20 schools. Our annual training of 150+ robotics teams has resulted in numerous champions, well-trained teams, and victories in 250+ competitions, reflecting our unwavering commitment to excellence.

10000+ Students at RFL Branch	**10** Private Labs Setup	**30+** Active B2B partnerships
30,000+ Students at Client Schools	**950+** Robotics Teams Trained	**250+** Competitions Won Globally

Top STEM Universities in USA

The competition training and exposure have played a pivotal role in shaping our students' profiles, paving the way for their admission to prestigious universities worldwide.



RFL Academy's Network of Robotics Labs

RFL Academy boasts a network of 10 state-of-the-art robotics labs across 5 cities in India and abroad. These cutting-edge facilities serve as more than just classrooms; they are innovation hubs for both regular courses and competition training, providing students with comprehensive educational experiences, an active learning environment, and hands-on experiences to fully engage students' minds.

MUMBAI

Juhu



Tardeo



Kandivali



AHMEDABAD

Satellite



South Bopal



SBR



SURAT

Vesu



Pal-Adajan



JAMNAGAR

Jamnagar



Panaji , Goa



Goa

Our School Partners

At RFL Academy, we're more than just a vendor; we are your implementation partner for all things tech with the help of Robotics, AI, Coding and Digital Literacy. Our curriculum also helps CBSE schools fulfil Kaushalya Bodh requirements for their school and ICSE schools to teach AI for Grades 9 and 11 as elective subject.

Robotics as a Service (RaaS) Model:

- **Expert Coaches:** Skilled coaches lead the tech curriculum within the school.
- **Tailored Curriculum:** Aligned with NEP 2020 for Indian schools and Common Core Standards for International schools.
- **Resources and Support:** We provide all resources and offer prep for competitions.

School Support (SS) Model:

- **Curriculum and Training:** We offer our thorough training for your teachers.
- **Ongoing Support & Investment Recommendations:** Incremental training sessions, regular observations, & feedbacks.
- **Competition Mentorship:** Guidance for mentoring school teams in competitions.

DHIRUBHAI AMBANI INTERNATIONAL SCHOOL ESTD. 2003

JBCN International School

THE HYDERABAD PUBLIC SCHOOL BEGUMPET

APPLE GLOBAL SCHOOL

UNIVERSAL

مدارس مسك Misk Schools

AMERICAN SCHOOL OF BOMBAY

Calorx Olive International School

GEMS GENESIS INTERNATIONAL SCHOOL MIND OPENERS

JBCN International School

LAKSHAYA INTERNATIONAL SCHOOL

SMT. KANCHANBEN BHIKHALAL SHAH TAPOVAN VIDYA VIHAR

THE H.B. KAPADIA NEW HIGH SCHOOL SINCE 1956 HBK

ARYA VIDYA MANDIR Group of Institutions

JAYSHREE PERIWAL INTERNATIONAL SCHOOL

And many more

Competitions RFLites have Championed

FIRST ROBOTICS COMPETITION

High school students build industrial-size robots under strict rules to compete in alliance games, fundraise, and promote STEM in their communities.

Total Teams Trained	National Championships	International Championships
2	6	4

FIRST TECH CHALLENGE

Students design, build, and code metal robots to compete in an alliance format using a Java-based programming platform powered by Android technology.

Total Teams Trained	National Championships	International Championships
20+	15+	5+

FIRST LEGO LEAGUE

Introduces children ages 4-16 to STEM through hands-on learning and robotics through Lego Education Kits, inspiring critical thinking, coding, and design skills.

Total Teams Trained	National Championships	International Championships
25+	10+	5+

WRO™
WORLD ROBOT OLYMPIAD

A global event founded in 2004, bringing together young people from over 85 countries to enhance their creativity and problem-solving skills through educational robot competitions using Lego Education as well as non Lego products across categories.

Total Teams Trained	National Championships	International Championships
350+	200+	5+

VEX IQ CHALLENGE
PITCHING IN

Provides elementary and middle school students with robotics challenges using VEX IQ kits that enhance their STEM skills through hands-on, student-centered learning.

Total Teams Trained	National Championships	International Championships
5	4	2

CODEAVOUR

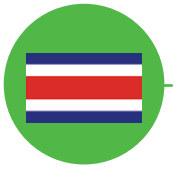
An annual global AI and coding competition for students aged 7-18, encouraging the creation of innovative projects in PictoBlox to develop critical 21st-century skills.

Total Teams Trained	National Championships	International Championships
100+	40+	10+

International Achievements

2017

WRO Lego Creativity Award, Costa Rica
(First year of launch)



2018

Best Robot Design Award at VEX World Championship, Kentucky, USA



2019

Trailblazing Triumphs, Rookie All Star Award at FIRST Robotics Competition, Detroit, USA



2020

Navigating Challenges, Victory at the FIRST Tech Challenge
Mexico (Remote)



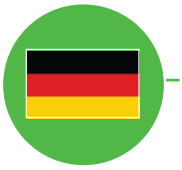
2021

First place in **solutions challenge** at **FIRST Global Challenge Second** in Robot Games, and the **Outstanding Mentor Award**



2022

Gold Medal at World Robot Olympiad



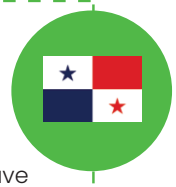
2023

Lab Fusion team from India secures the prestigious **Think Award** and claims **3rd place** overall at the **2023 FIRST World Championship**
USA



2023

- **Team Marine Bot** have clinched a **BRONZE World Robot Olympiad International**, Panama
- **FTC team sigma** 20890 **wins Inspire Award** at APOC, Sydney



2024

Rookie All Star award at **FIRST Robotics Championship**, USA



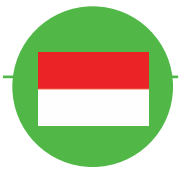
2025

- **Gujarat Tech Titans** secures **first place** in the Connect Award at the FTC Asia Pacific Open Championship in Sydney, Australia
- **Engineering Excellence Award** at FIRST World Championship, Houston, USA



2026

- **9692 Sigma clinched** Team Sustainability award **at FIRST world championship Houston**
- **CODEAVOUR 7.0 –** Team Lion Iron clinched **Gold at international finale,** Jakarta, Indonesia



Testimonials from School Leaders

JBCN International School

Our partnership with RFL has been incredibly successful. The team's promptness and the skills students have developed have truly impressed both us and the parents

Principal,
JBCN International School,
Mulund

HBK

Executing robotics at this scale was challenging, but RFL helped us build a curriculum that worked for every student.

Harsh Kapadia
Trustee, The H.B. Kapadia New High School

Scan to watch

APPLE GLOBAL SCHOOL

RFL's robotics program goes beyond maths and science. It builds problem solving skills that students carry with them for life.

Nivedita Chakravarti
Principal, Apple Global School

Scan to watch







PERSONAL INNOVATION PROJECTS

What is personal INNOVATION projects?

A personal innovation project is a real-world problem a student chooses to solve independently. With guidance from an RFL mentor, students take an idea from a rough sketch to a working prototype, building something that is truly their own.

Why It Matters

Personal innovation develops problem-solving, creativity, critical thinking, resilience, and ownership. More importantly, it gives students a tangible body of work that demonstrates initiative and real-world application. This builds students a tangible portfolio that demonstrates initiative and real-world problem-solving, strengthening their college applications, scholarship profiles, and academic portfolios.

The Process

Observe a problem

The student identifies something in their world that frustrates them, at home, at school, or in their city.

Imagine a solution

They sketch, argue, and prototype in cardboard before touching a single wire.

Build with guidance

An RFL mentor supports the build without taking it over. The student owns every decision.

Test and improve

The first version fails. It gets fixed. It fails again. Iteration becomes second nature.

Create real impact

Real-world impact transforms how students see their own potential.

Some of the the Personal Innovation Projects



Smart Earplug Headphone

Made by: Yashkit Jain & Sharanya · JPIS, Jaipur

Problem: Ear-pressure discomfort during flights, trekking and diving.

Solution: Pressure-balancing headphones with automated air equalization.

Outcome Deployed for IRIS Nationals



SOS Smart Shoes

Made by: Aahana Jain · Pune

Problem: Need for a discreet, hands-free way to call for help in emergencies.

Solution: Smart shoes with tap-based SOS, live GPS tracking and cellular alerts.

Outcome Deployed · 5-week build · ₹5,200 project cost



Multi-Factor Driver Authentication System

Made by: Viren Sood · JPIS, Jaipur

Problem: Vehicle security systems can be lost, copied or bypassed.

Solution: Dual biometric authentication using face recognition + fingerprint, with SMS-based remote control.

Outcome Deployed · ₹21,100 project cost

Scan this QR code to explore all the innovation projects built by our students.

COURSES WE OFFER

Our courses are aligned with NEP 2020 for Indian schools and Common Core Standards for international schools. Our curriculum is progressive in nature and allows students to explore each and every aspect of robotics Design, Build, Prototype, Code, Test & iterate with latest kits and Coding tools in hands on minds on manner, guided by our expert coaches.

Course Name	Grades
Foundational STEM Exploration	Grade 1
Simple Machines & Discovering Robotics	Grade 2
Core STEM Skills & Pre-Robotics	Grade 3
Programmable Robotics & AI Integration	Grade 4
Robotics & AI Tech Tinkers	Grade 5
Advanced Robotics & Applied AI	Grade 6
Electronics, Arduino & Robotics	Grade 7
IoT, Smart Systems & Applied Robotics	Grade 8
Engineering Robotics Course	Grade 9-12
School of AI	Grades 3 to 8



GRADE 1

FOUNDATIONAL STEM EXPLORATION



Young learners discover how everyday objects move, balance and work by building playful STEM models. Each activity turns a simple scientific idea into something they can see, touch and test. Introductory coding tasks help them understand sequences and instructions. The course builds curiosity, confidence and an early maker mindset.

Core learning areas:

- Motion, balance and structures
- Gears, levers and simple mechanisms
- Sequencing and introductory coding
- Creative model building

Platforms:

- LEGO Education STEM resources
- Beginner drag-and-drop coding tools

Skills developed:

- Fine motor skills
- Observation
- Sequential thinking
- Creative model building
- Creativity
- Teamwork

Competition pathway:

- Internal STEM showcases and school exhibitions.

Suggested models and projects:

- Pinwheel
- See-saw
- Spinning top
- Car launcher
- Aeroplane
- Crane
- Merry-go-round
- Candy dispenser

Includes Course Completion Certificate

GRADE 2

SIMPLE MACHINES & DISCOVERING ROBOTICS



Students move from static models to motorised creations that can react to simple inputs. They explore how machines transfer movement and how sensors make a model responsive. ScratchJr activities introduce events, loops and debugging through stories and games. The course strengthens cause-and-effect thinking and turns early curiosity into structured problem-solving.

Core learning areas:

- Simple machines and motorised motion
- Sensors and responsive models
- Events, loops and debugging
- Introductory AI interaction

Platforms:

- LEGO Simple Machines
- LEGO WeDo
- ScratchJr
- PictoBlox Jr.

Skills developed:

- Problem-solving
- Cause-and-effect understanding
- Coding logic
- Creativity
- Collaboration

Competition pathway:

- Internal STEM showcases and age-appropriate innovation events.

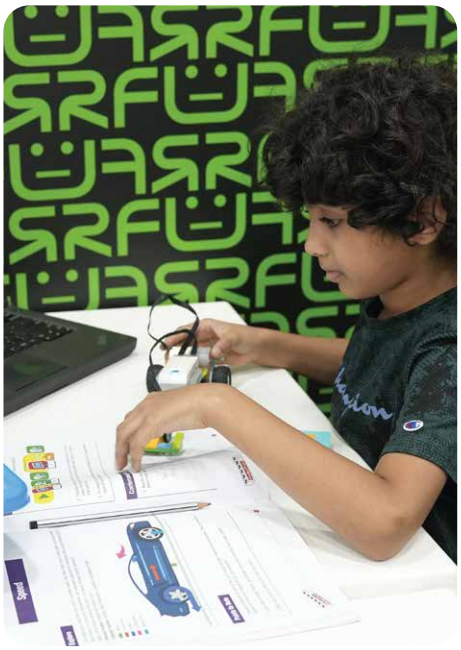
Suggested models and projects:

- Motorised carousel
- Sensor-based gate
- Moving animal model
- Interactive vehicle
- Animated story
- Hand-detection game

Includes Course Completion Certificate

GRADE 3

CORE STEM SKILLS & PRE-ROBOTICS



Students begin connecting mechanical principles with coding to create functional STEM projects. They investigate how gears, pulleys, levers and motorised systems can solve a task. Scratch and PictoBlox introduce algorithms, patterns and interactive programming. The course encourages students to think like young engineers: plan, build, test and improve.

Core learning areas:

- Simple machines and mechanical advantage
- Motorised mechanisms
- Scratch-based algorithms and patterns
- Build-test-improve design cycle

Platforms:

- LEGO Simple & Powered Machines
- LEGO WeDo
- ScratchJr
- PictoBlox Jr.

Skills developed:

- Engineering thinking
- Construction skills
- Computational thinking
- Logical sequencing
- Problem-solving

Competition pathway:

- Codeavour
- WSRO

Suggested models and projects:

- Balloon car
- Catapult
- Bucket-lifting mechanism
- Scissor lift
- Sweeper bot
- Tower crane

Includes Course Completion Certificate

GRADE 4

PROGRAMMABLE ROBOTICS & AI INTEGRATION



Students progress from motorised models to programmable robots that sense and respond to their surroundings. They use loops, conditions and sensor data to control movement and behaviour. Introductory computer vision and gesture-based activities show how AI can make robots more interactive. The course develops confidence in building, coding and debugging complete robotic systems.

Core learning areas:

- Robot construction and sensor integration
- Loops, conditions and responsive movement
- Line following and object detection
- Gesture control and introductory computer vision

Platforms:

- LEGO WeDo
- Quarky
- Scratch
- PictoBlox

Skills developed:

- Block-based programming
- Sensor integration
- Debugging
- Design thinking
- Creative problem solving

Competition pathway:

- Codeavour
- WSRO

Suggested models and projects:

- Line-following robot
- Object-detection robot
- Gesture-controlled robot
- Smart delivery robot
- AI expression project
- Sensor-based challenge robot

Includes Course Completion Certificate

GRADE 5
ROBOTICS & AI TECH TINKERS



Students enter structured robotics by building autonomous systems with LEGO Mindstorms EV3. They program motors and sensors to solve practical challenges involving navigation, detection and automation. PictoBlox introduces selected AI applications that complement the physical robot. The course creates a strong foundation in robotics programming and sensor-based decision-making.

Core learning areas:

- EV3 robot construction and programming
- Sensor-based automation
- Conditional logic and navigation
- Introductory AI applications

Platforms:

- LEGO Mindstorms EV3
- PictoBlox

Skills developed:

- Robotics programming
- Logical thinking
- Automation
- Sensor based decision making
- Teamwork

Competition pathway:

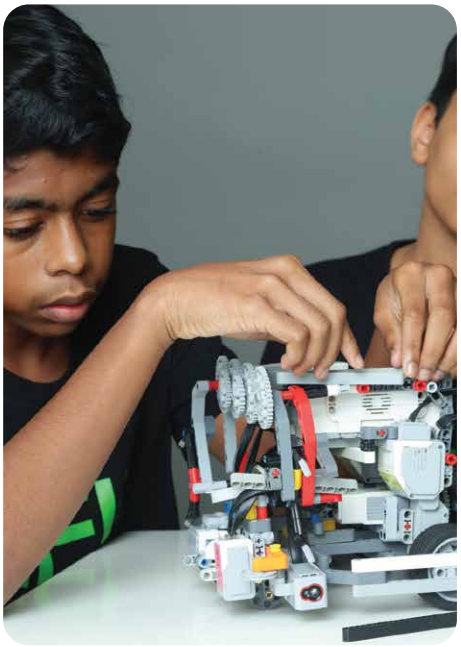
- WRO
- Codeavour
- WSRO
- VEX IQ

Suggested models and projects:

- Automated entrance door
- Mars rover
- Candy dispenser
- Line-following robot
- Smart streetlight
- Obstacle-detection robot

Includes Course Completion Certificate

GRADE 6
ADVANCED ROBOTICS & APPLIED AI



Students take on more complex challenges that require multiple sensors, autonomous navigation and stronger algorithms. They design robots that analyse inputs and make decisions without continuous human control. AI model training and computer vision expand the range of problems students can solve. The course emphasises experimentation, optimisation and dependable performance.

Core learning areas:

- Multi-sensor autonomous systems
- Navigation and maze-solving algorithms
- Computer vision and AI model training
- Testing and performance optimisation

Platforms:

- LEGO Mindstorms EV3
- Quarky
- PictoBlox

Skills developed:

- Algorithmic thinking
- Advanced programming
- AI model application
- Testing and optimisation
- Problem-solving

Competition pathway:

- WRO
- Codeavour
- VEX IQ
- WSRO

Suggested models and projects:

- Maze-solving robot
- Visitor counter
- Autonomous delivery robot
- Card-recognition system
- AI-controlled robot
- Smart monitoring system

Includes Course Completion Certificate

GRADE 7

ELECTRONICS, ARDUINO & ROBOTICS



Students transition from ready-made robotics platforms to electronic circuits and microcontrollers. They learn how components, sensors and code come together to create a working system. Arduino projects and circuit simulation provide a practical understanding of hardware-software integration. App development and introductory Python broaden their ability to control and communicate with devices.

Core learning areas:

- Electronic circuits and components
- Arduino programming and sensor control
- Circuit simulation and robotics
- App development and introductory Python

Platforms:

- Arduino Uno
- Tinkercad
- MIT App Inventor
- PictoBlox

Skills developed:

- Circuit building
- Hardware-software integration
- Microcontroller programming
- App development
- Analytical thinking

Competition pathway:

- FTC
- WRO
- Codeavour
- WSRO

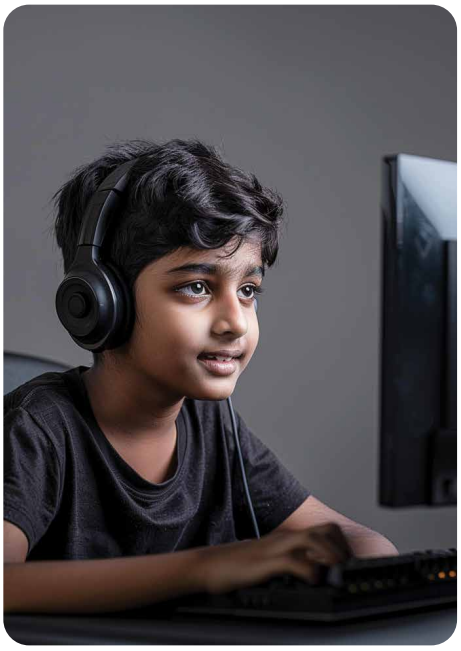
Suggested models and projects:

- Water-level indicator
- Automatic streetlight
- Obstacle-avoiding robot
- Line-following robot
- Security alarm
- Weather monitoring system

Includes Course Completion Certificate

GRADE 8

IoT, SMART SYSTEMS & APPLIED ROBOTICS



Students design connected systems that collect data, communicate online and automate real-world tasks. Using ESP32, sensors and cloud platforms, they move from standalone devices to complete IoT solutions. Arduino C and Python introduce text-based programming for more advanced control. The course prepares students to develop practical, connected and user-focused technology products.

Core learning areas:

- ESP32 and IoT connectivity
- Sensors, cloud data and automation
- Arduino C and Python programming
- Smart-system product development

Platforms:

- ESP32
- Arduino C
- Python
- IoT cloud platforms

Skills developed:

- IoT system design
- Text-based programming
- Data handling
- Automation
- Product development

Competition pathway:

- FTC
- WRO
- Codeavour
- WSRO

Suggested models and projects:

- IoT weather assistant
- Smart parking indicator
- Home automation system
- Voice-controlled lights
- Smart security system
- Connected environmental monitor

Includes Course Completion Certificate

GRADES 9 TO 12

ENGINEERING ROBOTICS COURSE



Students move beyond educational kits and begin working with metal robots, engineering mechanisms and fabrication tools. The course combines CAD, workshop practices, robot programming and 3D printing in one practical engineering journey. Students design, build and test mechanisms such as drivetrains, grabbers, shooters and lifting systems. It prepares them to contribute meaningfully to advanced robotics teams and engineering projects.

Core learning areas:

- Metal robot construction and control systems
- CAD, engineering drawing and 3D printing
- Drivetrains, manipulators and advanced mechanisms
- Autonomous control, vision and systems integration

Platforms:

- Metal robotics construction system
- Onshape
- 3D printers
- Workshop tools
- FIRST-style robot control systems

Skills developed:

- IoT system design
- Text-based programming
- Data handling
- Automation
- Product development

Competition pathway:

- IRIS
- WRO
- FRC
- FTC
- Codeavour

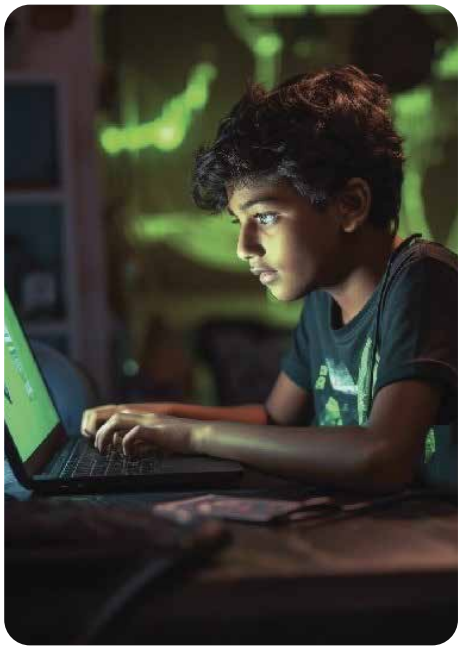
Suggested models and projects:

- Mecanum drive robot
- Base robot with grabber
- CAD robot assembly
- Flywheel shooting mechanism
- Hanging robot
- Pick-and-place robot
- 3D-printed functional components

Includes Course Completion Certificate

GRADES 3 TO 8

SCHOOL OF AI + COMPUTATIONAL THINKING



This structured six-year programme helps students understand, question and build with Artificial Intelligence, with a curriculum aligned to CBSE learning guidelines. Computational Thinking develops the logic, patterns and problem-solving habits that form the foundation of AI literacy.

Students progress from recognising AI in everyday life to training models and creating purposeful digital products. Equal emphasis is placed on responsible use, verification, privacy and knowing how to use AI thoughtfully and ethically.

Programme structure:

- 10 School of AI sessions per grade
- 5 Computational Thinking sessions per grade
- 90 total student sessions
- 10+ real AI tools
- Paper-based CT worksheets and digital AI projects

Grade progression:

- Grade 3: Explorers
- Grade 4: Detectives
- Grade 5: Creators
- Grade 6: Navigators
- Grade 7: Builders
- Grade 8: Innovators

Core learning areas:

- Patterns, algorithms and decomposition
- Logic, variables and optimisation
- AI literacy, model training and digital products

Includes Course Completion Certificate

KICKSTART YOUR CHILD'S JOURNEY IN ROBOTICS AND AI FROM GRADE 1

Grade 1
Foundational STEM Exploration

Grade 2
Simple Machines & Discovering Robotics

Grade 3
Core STEM Skills & Pre-Robotics

School of AI + Computational Thinking

Grade 4
Programmable Robotics & AI Integration

Grade 5
Robotics & AI Tech Tinkers

Grade 6
Advanced Robotics & Applied AI

Grade 7
Electronics, Arduino & Robotics

Grade 8
IOT, Smart Systems & Applied Robotics

Grade 9-12
Engineering Robotics Course

Scan for Details